

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes polymerization and synthesis data from six key patents focusing on Low Acrylonitrile Nitrile Butadiene Rubber (Low ACN NBR). The scope of this research covers the production of nitrile rubbers with bound acrylonitrile contents typically ranging from 27.9% to 36.6%. The selected patent landscape reveals two primary technical focuses: first, low-temperature (7°C to 8°C) emulsion polymerization processes utilizing redox initiator systems (such as diisopropylbenzene hydroperoxide or p-menthane hydroperoxide combined with iron(II) salts) and incremental chain-transfer agent (TDM) dosing to control molecular weight and Mooney viscosity; second, highly controlled post-polymerization coagulation and washing processes. A significant portion of the patent art focuses on adjusting latex pH and selecting specific coagulation salts (magnesium chloride, calcium chloride, or sodium chloride) to regulate the residual cation profile (ion index) of the final rubber. This precise ion control is shown to directly optimize vulcanization kinetics, reduce Mooney scorch, and improve the corrosion resistance and aging properties of the resulting vulcanizates.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

Patent 1

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2017 - Polymer Type: Low Acrylonitrile Nitrile Butadiene Rubber (NBR) - Relevance to target: Discloses a low-temperature emulsion polymerization process for NBR with low bound ACN (27.9% - 28.1%) using incremental additions of t-dodecyl mercaptan (TDM) to control Mooney viscosity.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization - Monomer system: Acrylonitrile and Butadiene - Monomer ratio: 500 g ACN / 4.37 kg Butadiene (approx. 10.3 wt% ACN in feed) or 815 g ACN / 4.0 kg Butadiene (approx. 16.9 wt% ACN in feed) - Water amount: 9.92 kg deionized water - Emulsifier: Potassium stearate and DNMK (disproportionated rosin acid potassium salt) - Emulsifier loading: 0.53 kg of 5.4 wt% potassium stearate solution, 0.440 kg DNMK - Initiator: Diisopropylbenzene hydroperoxide (DIHP) and Sodium formaldehyde sulfoxylate (FSS) - Initiator loading: 6 g DIHP, 139 g FSS aqueous solution - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: t-Dodecyl mercaptan (TDM) - Chain-transfer dosage: 18.72 g initially, with booster additions of

2.34 g at 37% and 59% conversion (or 21.85 g initially and 5.45 g booster) - Polymerization temperature: 7°C - Pressure: Not disclosed in the analyzed patent evidence - pH: Latex pH of 8.5 - 8.6 - Reaction time: 7 hours - Conversion: 70% (stopped with hydroxylamine sulfate) - Coagulation conditions: Coagulated using magnesium sulfate (4 kg saturated solution in 43 kg water) at elevated temperature, followed by washing at pH 11.5-11.8 (using KOH) - Post-treatment: Filtered and dried in an air oven at 100°C for 24 hours (residual humidity < 1%) - Raw polymer properties: Bound ACN content of 27.9% - 28.1% (Latex A/C) or 33.9% (Latex B); Mooney viscosity of 29.0 - 60.5 (ML 1+4 @ 100°C)

Relevant Experimental Evidence - Latex A was prepared with 500 g ACN, 4.37 kg butadiene, and 18.72 g TDM, yielding 28.1% bound ACN and 60.5 Mooney viscosity. - Latex C was prepared similarly to Latex A, yielding 27.9% bound ACN and 29.0 Mooney viscosity. - Latex B was prepared with 815 g ACN, 4.0 kg butadiene, and 21.85 g TDM, yielding 33.9% bound ACN and 32.1 Mooney viscosity. - Coagulation was performed using magnesium sulfate, followed by washing at pH 11.5-11.8 with KOH to control residual emulsifier and ash content.

Technical Relevance Demonstrates a low-temperature (7°C) emulsion polymerization process for low-ACN nitrile rubbers (27.9-28.1% bound ACN) using incremental additions of TDM to control molecular weight and Mooney viscosity.

Patent 2

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2015 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Focuses on the post-polymerization processing, latex coagulation, and vulcanization of nitrile rubbers to optimize aging and corrosion properties.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization (implied by latex processing) - Monomer system: Not disclosed in the analyzed patent evidence - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: Not disclosed in the analyzed patent evidence - pH: Not disclosed in the analyzed patent evidence - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: Not disclosed in the analyzed patent evidence - Coagulation conditions: Coagulation of 25 L of latex using magnesium chloride and an acid-ashed gelatin (viscosity 92.1 cP, IEP 8.7 in 10% aqueous solution) - Post-treatment: Vulcanization at 160°C under 120 bar hydraulic pressure for 30 minutes - Raw polymer properties: Mooney sheets subjected to hot air aging at 100°C for 48 hours; vulcanizates evaluated for stress at 300% elongation, tensile strength, and elongation at break

Relevant Experimental Evidence - Latex A (25 L) was coagulated using a magnesium chloride solution containing dissolved acid-ashed gelatin (10% strength aqueous solution, viscosity 92.1 cP, IEP 8.7). - Vulcanized test specimens were prepared at 160°C under 120 bar pressure for 30 minutes. - Corrosion testing of sealing rings was conducted between sand-blasted aluminum plates at 80% relative humidity and 70°C.

Technical Relevance Highlights the use of specific coagulation aids (gelatin in magnesium chloride solution) to control the ion index and improve the vulcanization and corrosion properties of NBR.

Patent 3

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2011 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Part of a family of patents addressing the synthesis and ion-index control of nitrile rubbers.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization (implied) - Monomer system: Not disclosed in the analyzed patent evidence - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: Not disclosed in the analyzed patent evidence - pH: Not disclosed in the analyzed patent evidence - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: Not disclosed in the analyzed patent evidence - Coagulation conditions: Not disclosed in the analyzed patent evidence - Post-treatment: Not disclosed in the analyzed patent evidence - Raw polymer properties: Not disclosed in the analyzed patent evidence

Relevant Experimental Evidence - No specific experimental parameters are disclosed in the provided text for this patent.

Technical Relevance Represents the foundational patent family for controlling the cation concentration and ion ratios in nitrile rubbers to optimize vulcanization kinetics.

Patent 4

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Investigates the effect of coagulation salt type (e.g., CaCl₂ vs. MgCl₂) and cation concentration on Mooney scorch, vulcanization rate, and mechanical properties of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization (implied by latex coagulation) - Monomer system: Not disclosed in the analyzed patent evidence - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: Not disclosed in the analyzed patent evidence - pH: Not disclosed in the analyzed patent evidence - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: Not disclosed in the analyzed patent evidence - Coagulation conditions: Coagulated using calcium chloride (CaCl₂) or magnesium chloride (MgCl₂) - Post-treatment: Vulcanization at 160°C under 120 bar hydraulic pressure for 30 minutes - Raw polymer properties: Mooney scorch, vulcanization rate (t₉₀ - t₁₀), and stress at 300% elongation (7.0 - 10.2 MPa)

Relevant Experimental Evidence - Nitrile rubbers were processed in a 1.5 L mechanical laboratory mixer using 100 parts by weight of NBR. - Coagulation with CaCl₂ resulted in specific cation concentrations (e.g., 1190-1290 ppm Ca) and vulcanizate properties (stress at 300% elongation of 7.0-10.2 MPa, elongation at break of 608-651%). - Coagulation with magnesium chloride yielded latices with lower Mooney scorch, shorter vulcanization times, and higher stress at 300% elongation compared to non-inventive rubbers.

Technical Relevance Demonstrates that the choice of coagulation salt (MgCl₂ vs. CaCl₂) and the resulting cation ratio in the rubber significantly influence Mooney scorch and vulcanization rates.

Patent 5

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Butadiene Rubber (NBR) / Carboxylated NBR (XNBR) - Relevance to target: Discloses a low-temperature emulsion polymerization process using a specific redox initiator system (Trigonox NT 50 and Fe(II) chloride or sulfate) to produce NBR with controlled ACN content and Mooney viscosity.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization - Monomer system: Butadiene, Acrylonitrile, and 2-Hydroxyethyl methacrylate (HEMA) - Monomer ratio: Butadiene: 61.4, Acrylonitrile: 36, HEMA: 2.6 parts by weight - Water amount: 220 parts by weight (total water); 2.1 kg water for 1.25 kg monomer autoclave batches - Emulsifier: Sodium lauryl sulphate (Texapon K-12) - Emulsifier loading: 3.0 parts by weight (range 0.2 to 15 parts by weight) - Initiator: p-Menthane hydroperoxide (Trigonox NT 50) and Fe(II)SO₄ or Fe(II)Cl₂ redox system with Rongalit C - Initiator loading: 0.02 parts by weight Trigonox NT 50; Premix containing 0.986 g Fe(II)SO₄·7H₂O or 0.845 g Fe(II)Cl₂·4H₂O and 2.0 g Rongalit C in 400 g water - Catalyst / activator: EDTA (equimolar to Fe(II)), Na₂SO₄ (0.12 parts by weight) - Chain-transfer agent: t-DDM (tertiary dodecyl mercaptan) - Chain-transfer dosage: 0.54 parts by weight - Polymerization

temperature: $8.0^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$ - Pressure: 50 bar to 150 bar (for hydrogenation, if applicable) - pH: 7.0 ± 1.0 - Reaction time: 6.5 to 7.0 hours - Conversion: 84.2% to 86.1% (stopped with diethylhydroxylamine) - Coagulation conditions: Latex admixed with 50% dispersion of Vulkanox BKF (0.3 wt% on NBR solids) before coagulation - Post-treatment: Stabilized with Wingstay 29/Naugawhite dispersion (0.18 parts by weight) - Raw polymer properties: Bound ACN content of 35.3% - 36.6%; Mooney viscosity of 25 - 142 MU (ML 1+4 @ 100°C); Tg of -23.1°C to -25.8°C

Relevant Experimental Evidence - Nitrile rubber G (using FeSO_4 premix) achieved 86.1% conversion in 7.0 hours, yielding a polymer with 36% ACN. - Nitrile rubber H (using FeCl_2 premix) achieved 84.2% conversion in 6.5 hours. - The use of FeCl_2 instead of FeSO_4 in the redox system resulted in distinct polymer properties, showing the sensitivity of the redox catalyst system.

Technical Relevance Provides a detailed recipe for low-temperature emulsion polymerization of NBR with ~36% ACN using a p-menthane hydroperoxide/ Fe(II) redox system to control conversion and molecular weight.

Patent 6

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2014 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Investigates the impact of latex pH adjustment and coagulation salt type (NaCl vs. CaCl_2) on the ion index and subsequent vulcanization rate of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization (implied by latex processing) - Monomer system: Not disclosed in the analyzed patent evidence - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: Not disclosed in the analyzed patent evidence - pH: Latex pH adjusted to 5.8 (using HCl) or 8.4 - 9.5 (using KOH) before coagulation - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: Not disclosed in the analyzed patent evidence - Coagulation conditions: Coagulated using NaCl (15-26 wt% salt) or CaCl_2 (0.3-1.2 wt% salt) at temperatures of 20°C to 50°C - Post-treatment: Vulcanized in a 1.5 L laboratory mixer and evaluated for vulcanization rate (t_{90} - t_{10}) and mechanical properties - Raw polymer properties: Ion indices (Ca, Na, Mg, K levels); vulcanizates show stress at 30% elongation of 4.2 - 10.2 MPa and elongation at break of 535 - 651%

Relevant Experimental Evidence - Latex pH was adjusted using 5% aqueous KOH or 5% aqueous HCl before coagulation. - Coagulation with NaCl (e.g., examples C6-C9) required higher salt concentrations (15-26 wt%) and resulted in higher residual sodium levels (up to 960 ppm). -

Coagulation with CaCl₂ (e.g., examples C10-C13) required much lower salt concentrations (0.3-1.2 wt%) and resulted in high calcium levels (~1200 ppm) and higher vulcanization rates.

Technical Relevance Demonstrates how pre-coagulation pH adjustment and salt selection (NaCl vs. CaCl₂) control the residual ion profile, directly affecting the vulcanization rate and mechanical performance.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No cross-patent comparison provided.

4. REFERENCES

#	Patent No.	Title	Assignee	Jurisdiction	Year	Source
1	US7923518	Nitrile rubbers	Not disclosed	US	2011-04-12	Google Patents
2	US8389623	Nitrile rubbers	Not disclosed	US	2013-03-05	Google Patents
3	US8436091	Nitrile rubbers	Not disclosed	US	2013-05-07	Google Patents
4	US8664340	Nitrile rubbers	Not disclosed	US	2014-03-04	Google Patents

5	US9034992	Nitrile rubbers which optionally contain alkylthio terminal groups and which ...	Not disclosed	US	2015-05-19	Google Patents
6	US9650502	Process for the preparation of nitrile rubbers	Not disclosed	US	2017-05-16	Google Patents